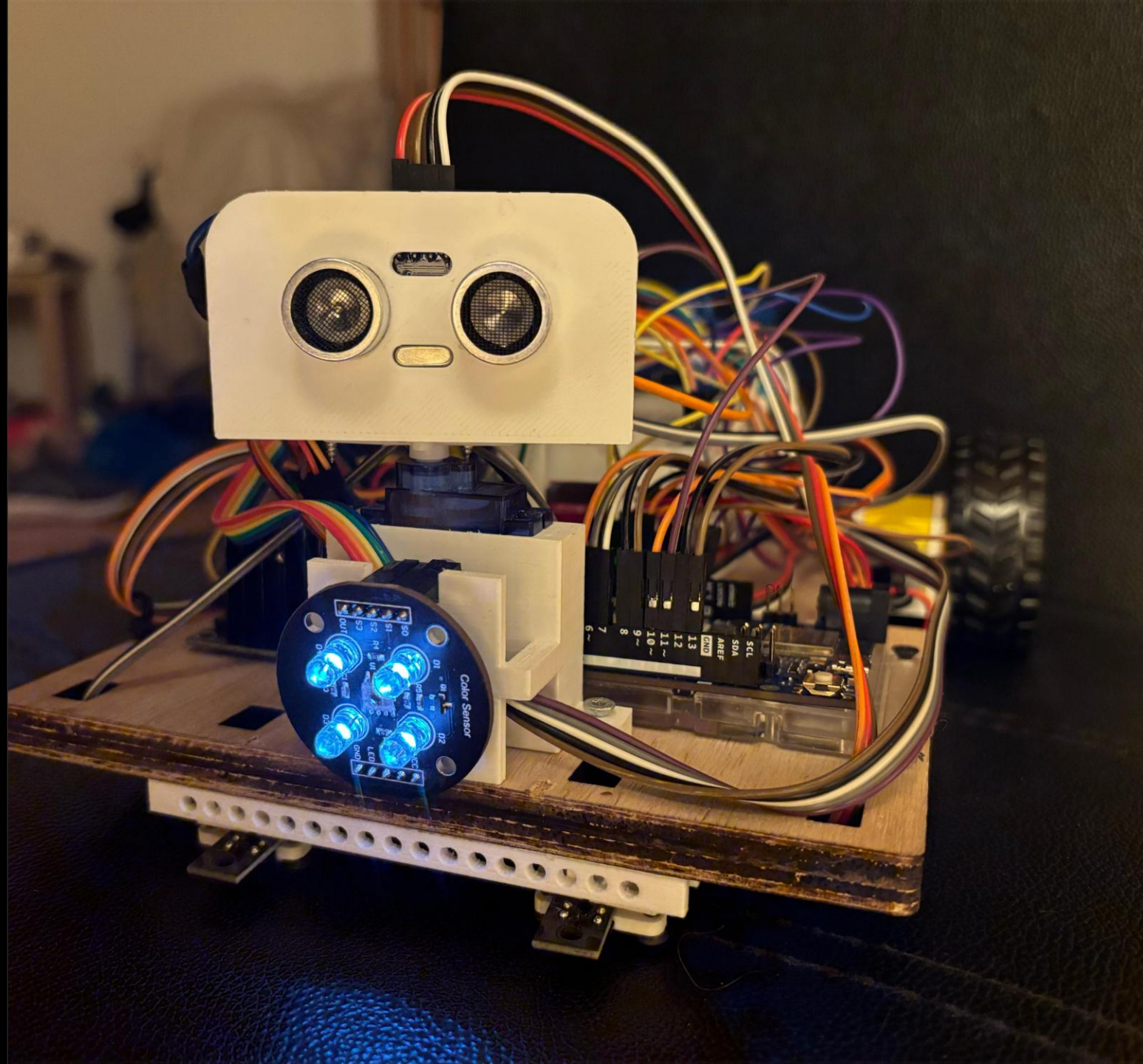
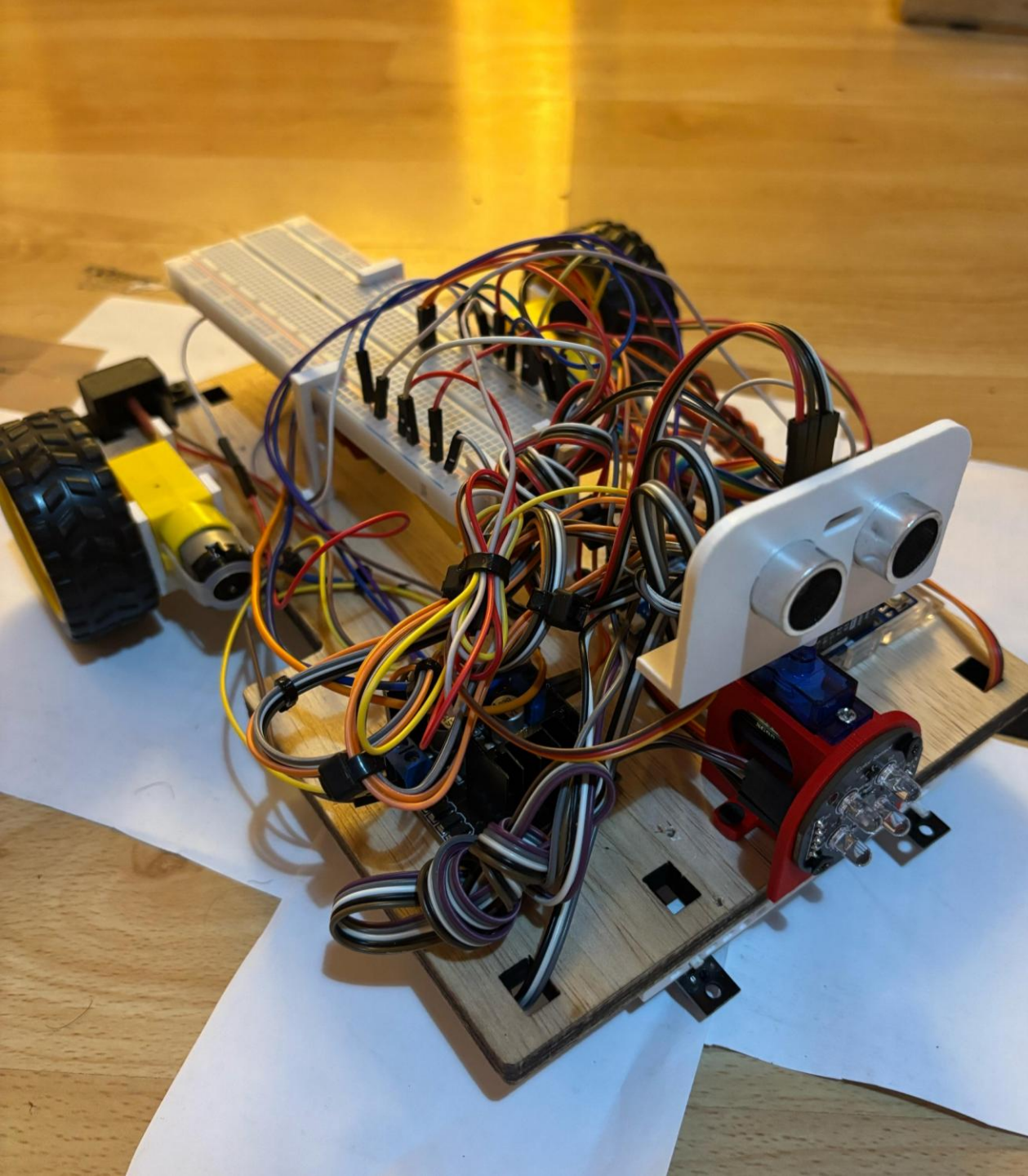

SYSTEM ENGINEERING & PROTOTYPING

Team: C1

Members:

- Md Mostafizur Rahman
- Mohamed Abdo
- Md Ruhul Kuddus Saleh
- M Amir Ejaz

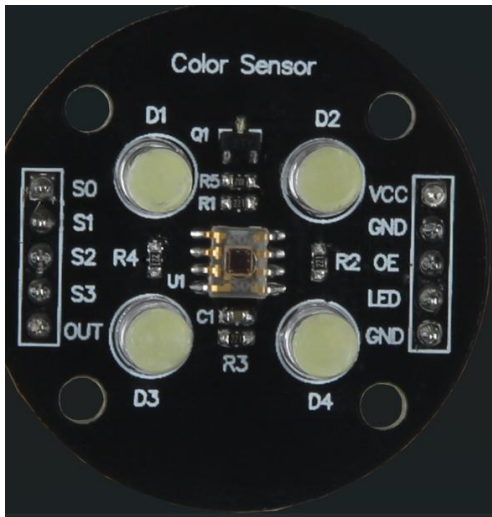
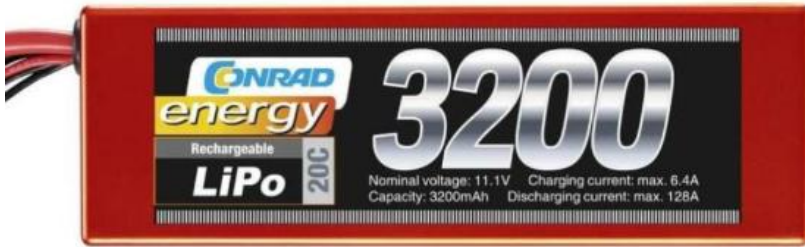




SYSTEM OVERVIEW

- Developing an autonomous car using Arduino.
 - Integrates ultrasonic sensors for obstacle detection
 - Infrared sensors for line following
 - Uses color sensor for color detection
 - DC motors drive the car.
 - Servo motor for obstacle removal
-

COMPONENTS



Robotic_Car_TeamC1-HSHL Public

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main 1 Branch 0 Tags

Go to file Add file Code

AamirEjaz SysML 675b5b5 · 2 hours ago 82 Commits

Color-Servo-Ultra-IR	Update Color-Servo-Ultra-IR.ino	last week
Documentation pictures	Updated SysML	6 hours ago
Ejaz-Code	Ejaz-Code	last week
Last_Updates_By_Abdo	Last_Code_Updates_&_Edited_PID	last week
Main_Components	Milestone1_by_Abdo	2 months ago
MileStone4	circuitmultisim	last month
Milestone_1	Milestone_Update	2 months ago
Milestone_1_Update	U-S_&_UR_holders_Update	2 months ago
Presentation SysML	SysML	2 hours ago
Print	print	last month
Rahman_Updates	Create Lane_detection_and_COLOR_work_by_rahman.ino	3 hours ago
Ruhul_Updates	Last_Code_Updates_&_Edited_PID	last week
SolidWorks_Componets_by_Abdo	SysML-Group_C1-update-by_Ejaz	last month
Task-1	Prototype Team C1 Task Updates	last week
Task-2	Prototype Team C1 Task Updates	last week
Task-3	Prototype Team C1 Task Updates	last week
Task-4	Prototype Team C1 Task Updates	last week
Task-5	Prototype Team C1 Task Updates	last week
Task-6	Prototype Team C1 Task Updates	last week
Task-7	Last_Code_Updates_&_Edited_PID	last week
Updated SysML	Updated SysML	6 hours ago
###_Final_Code_In_Sha'_Allah_###.cpp	Add files via upload	3 days ago
Front_View_Car.png	Milestone1_by_Abdo	2 months ago
How To use GIT.pdf	update pdf	2 months ago
LICENSE	Initial commit	2 months ago
Last_Update.ino	Add files via upload	last week

About

Autonomous Vehicle Project Goal: Build a self-driving vehicle using line and obstacle detection, color-based navigation, and route optimization. Tasks include oval/figure-eight driving, parking, and speed tuning. Team shares all code, CAD files, and progress via Git. Tasks tracked in 'Tasks.xlsx'.

Readme GPL-3.0 license Activity 2 stars 2 watching 0 forks Report repository

Releases

No releases published [Create a new release](#)

Packages

No packages published [Publish your first package](#)

Contributors 4

MoAbdoHSHL Mohamed Abdo RK-Saleh sohan2961 Md Mostafizur Rahman AamirEjaz

Languages

C++ 100.0%

Suggested workflows

Based on your tech stack

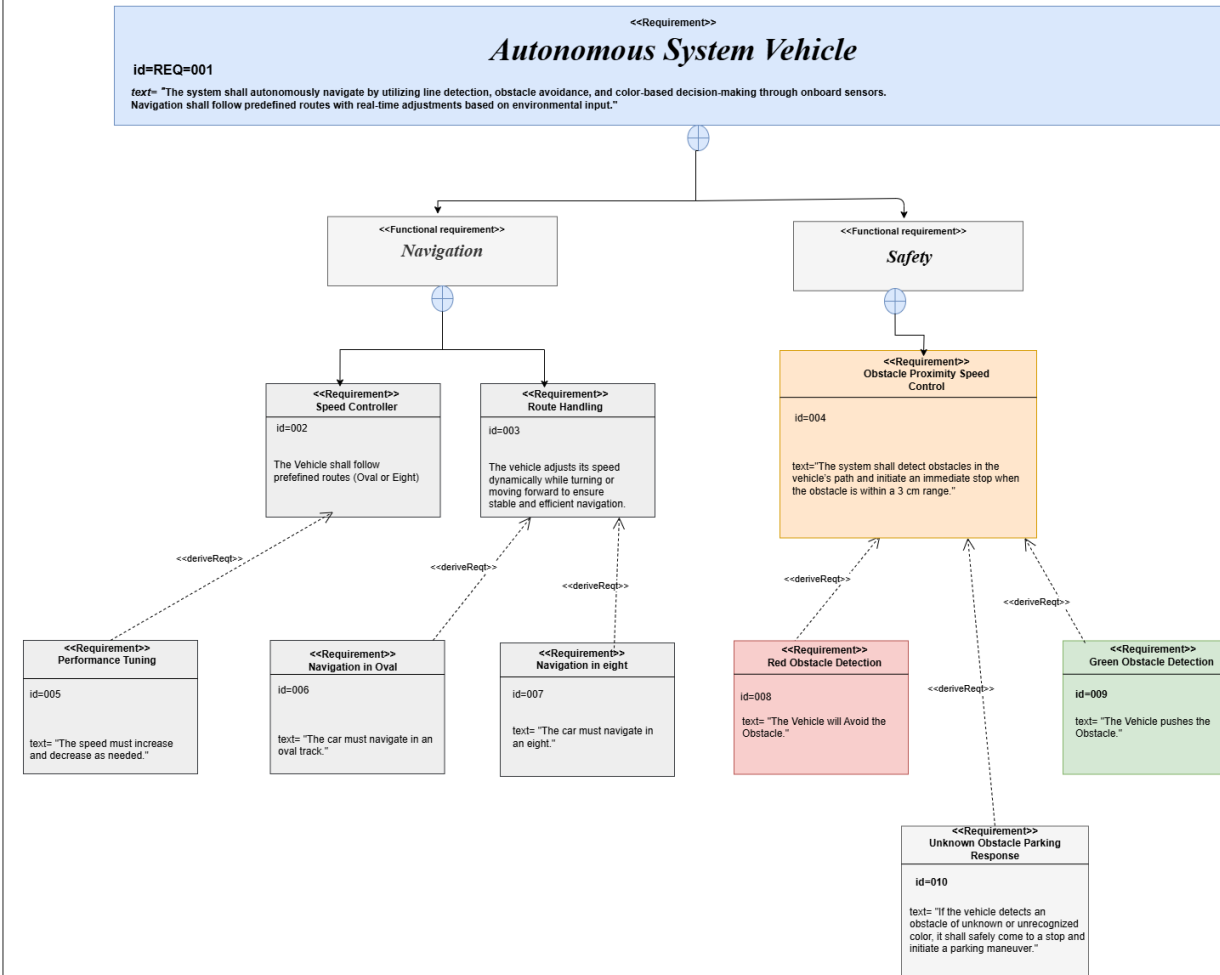
C/C++ with Make Configure

Build and test a C/C++ project using Make

TEAM CONTRIBUTION & GIT COMMITS

https://github.com/MoAbdoHSHL/Robotic_Car_TeamC1-HSHL

req [Autonomous Path-Following and Obstacle Detection System]

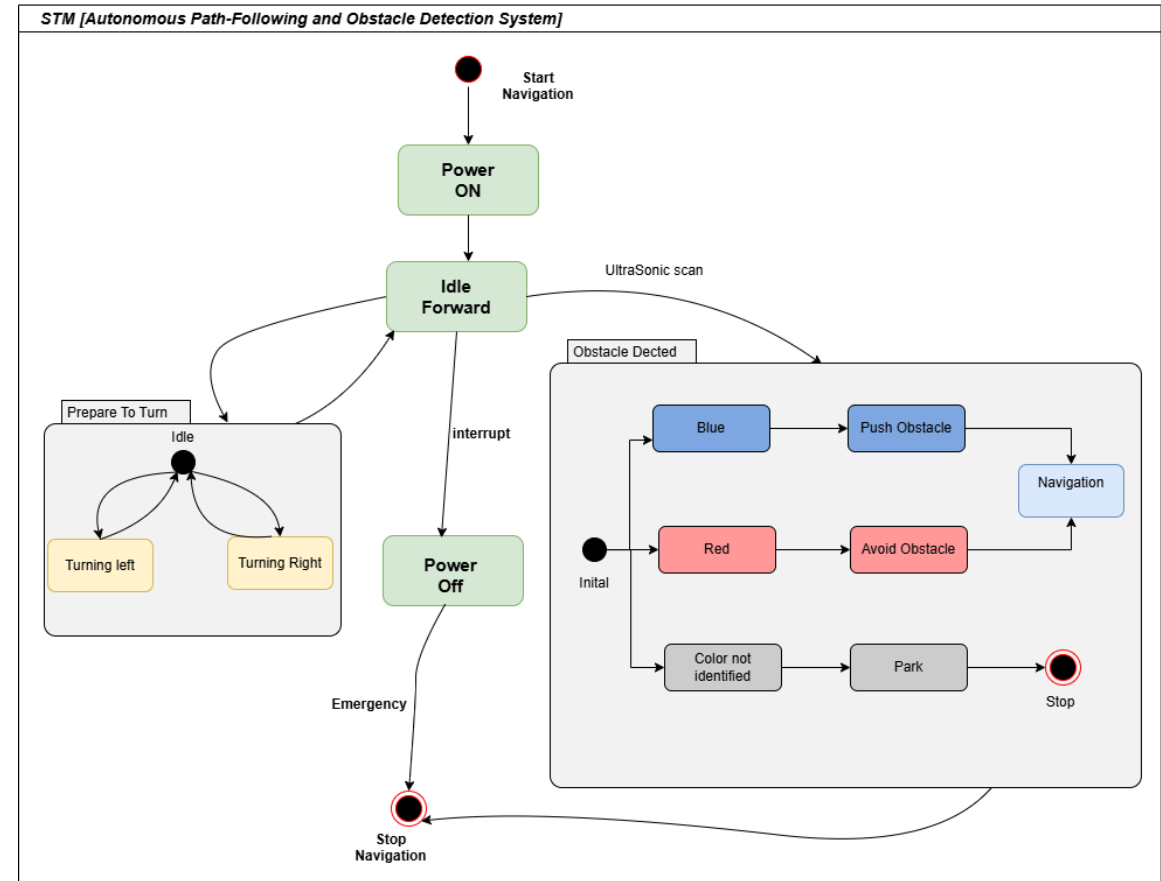


Requirement

- Follow a lane
- Drive different routes
- Optimize Voltage and Speed
- Detect Obstacles
- Detect Different color and take actions
- Push / Overtake an obstacle
- Park the car

STATE MACHINE DIAGRAM

- Starts with "Power ON,,
- Moves to "Idle Forward,,
- Detects obstacles and color.
- Responds (push, avoid, or park).
- Handles turning, interrupts, and emergencies.



AD [Autonomous Path-Following and Obstacle Detection System]

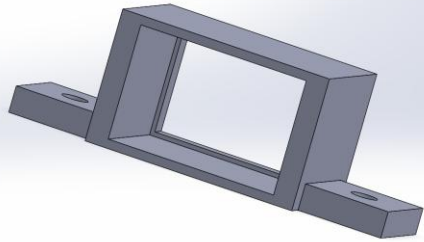


ACTIVITY DIAGRAM

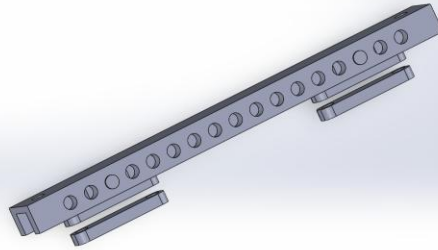
- Starts with autonomous navigation.
- Decides to move left, right, or forward.
- Checks if it's on the path.
- Detects obstacles & responds (avoid, push, or stop).
- Ends by stopping or parking.

3D PRINTED PARTS

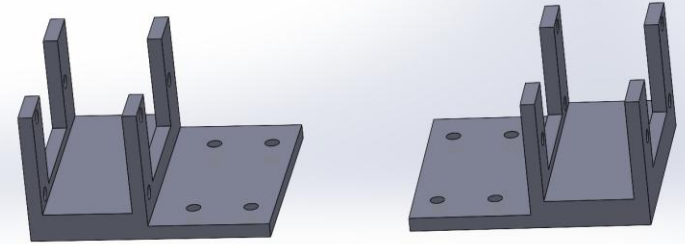
Switch Holder



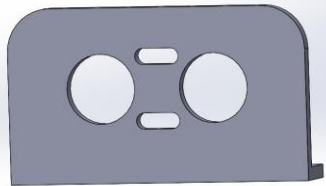
IR Holder



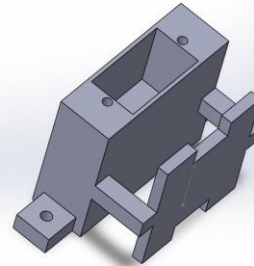
Motor Holder



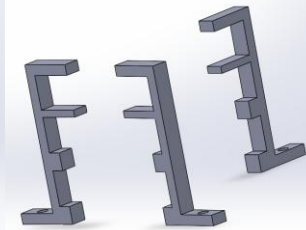
Ultrasonic Holder



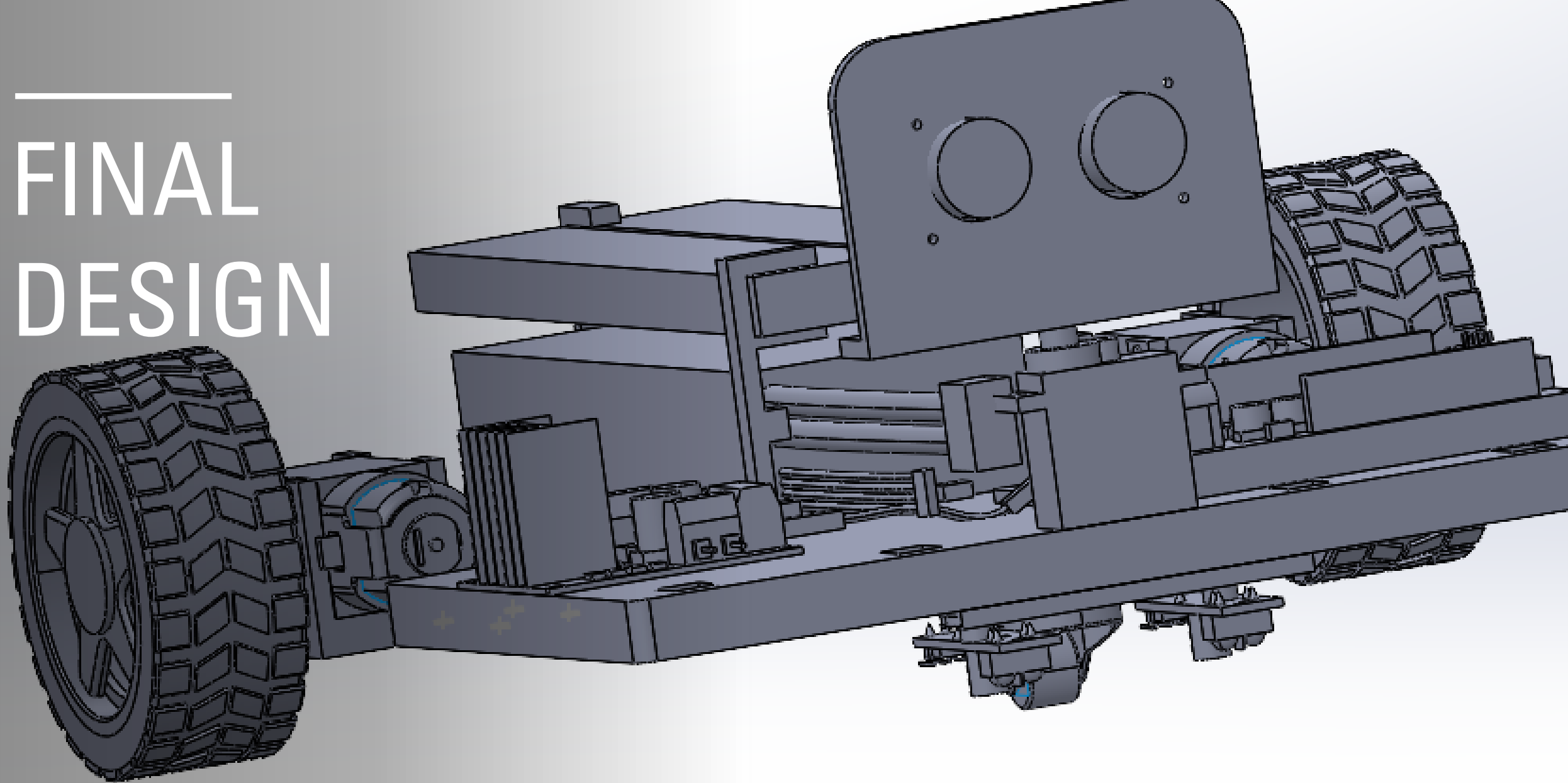
Color & Servo Holder

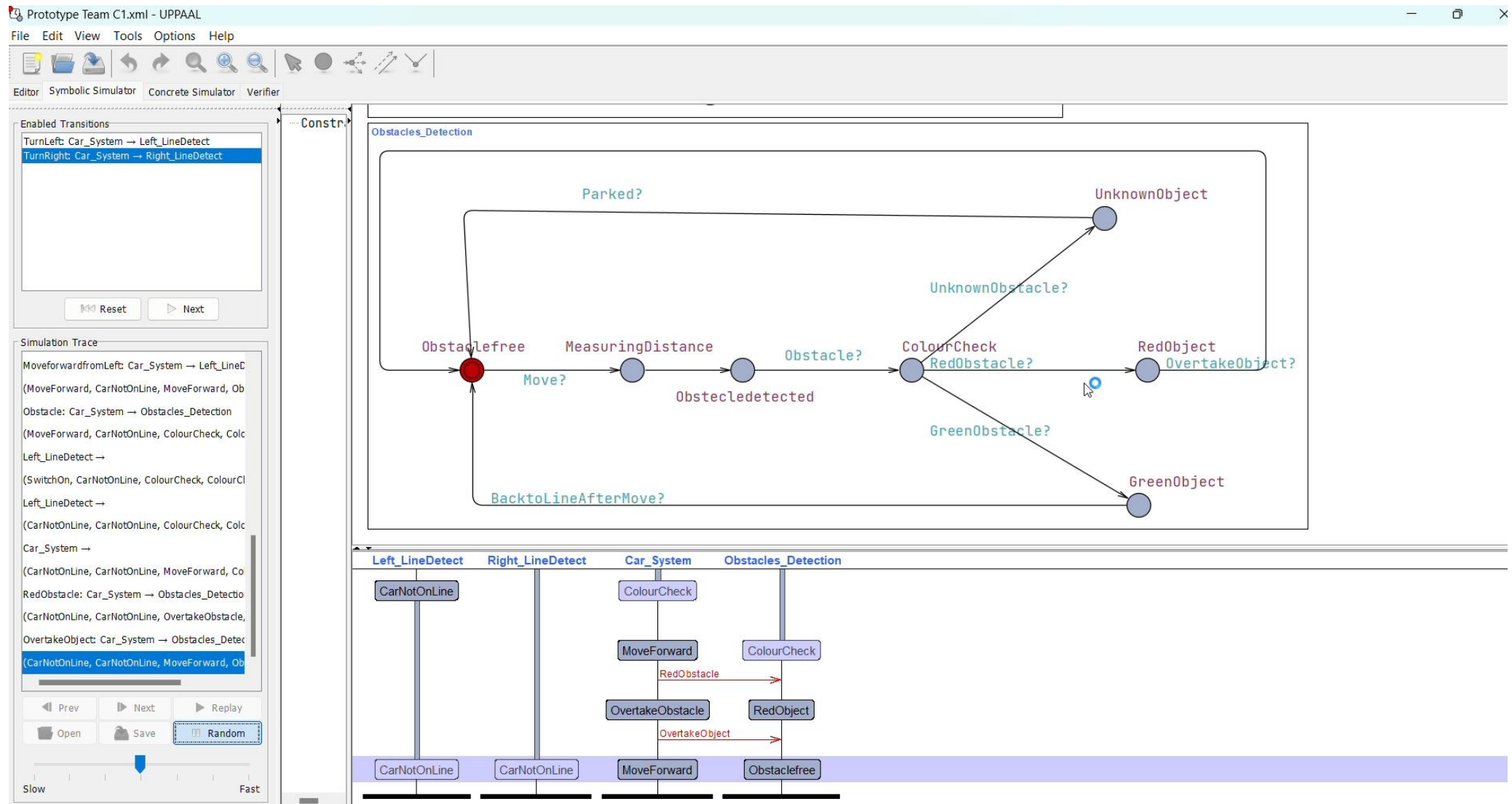


Battery Holder



FINAL DESIGN

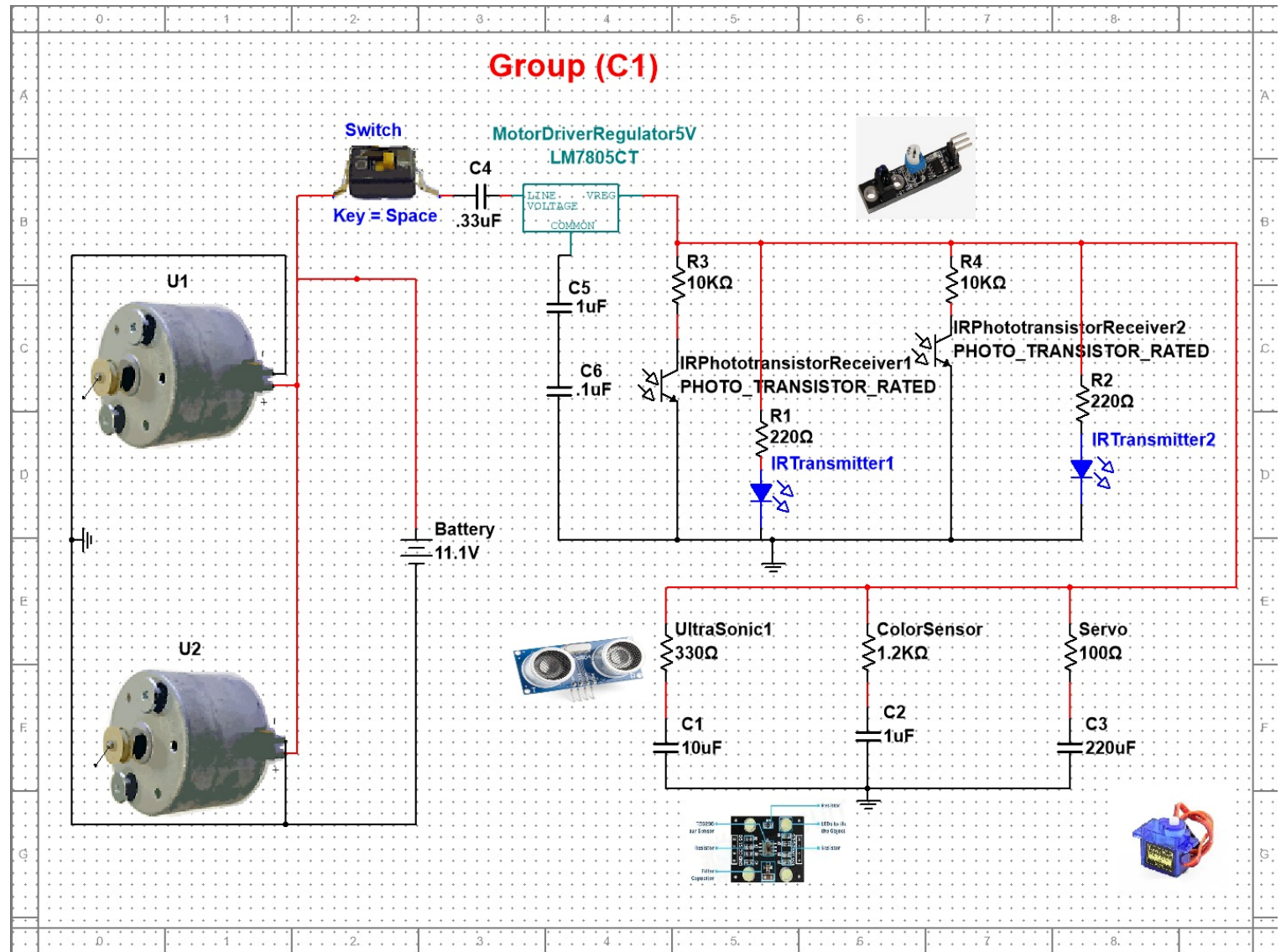




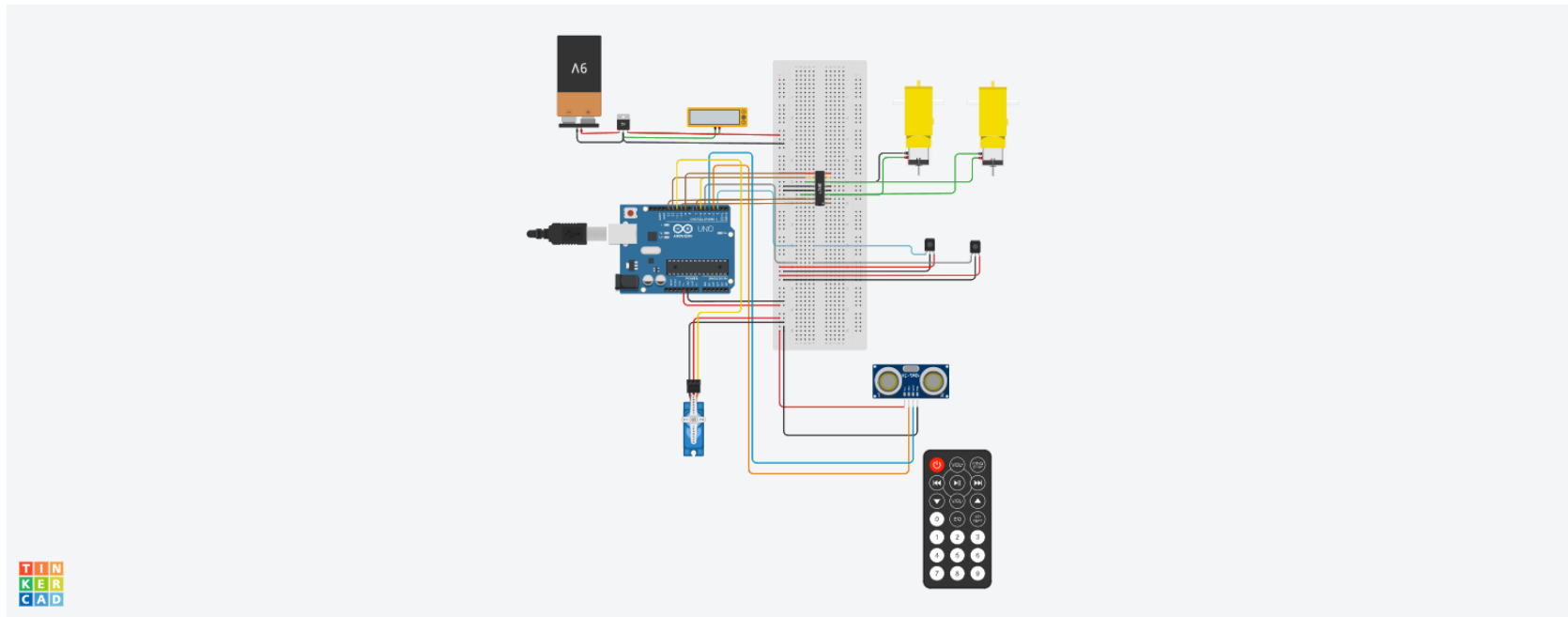
UPPAL SIMULATION

VOLTAGE & CURRENT MEASUREMENT

- Uses motors, sensors, and a battery.
- voltage regulator and capacitors.
- IR and color sensors detect obstacles.
- Power flows through regulated paths.



SCHEMATIC VIEW ON TINKERCAD

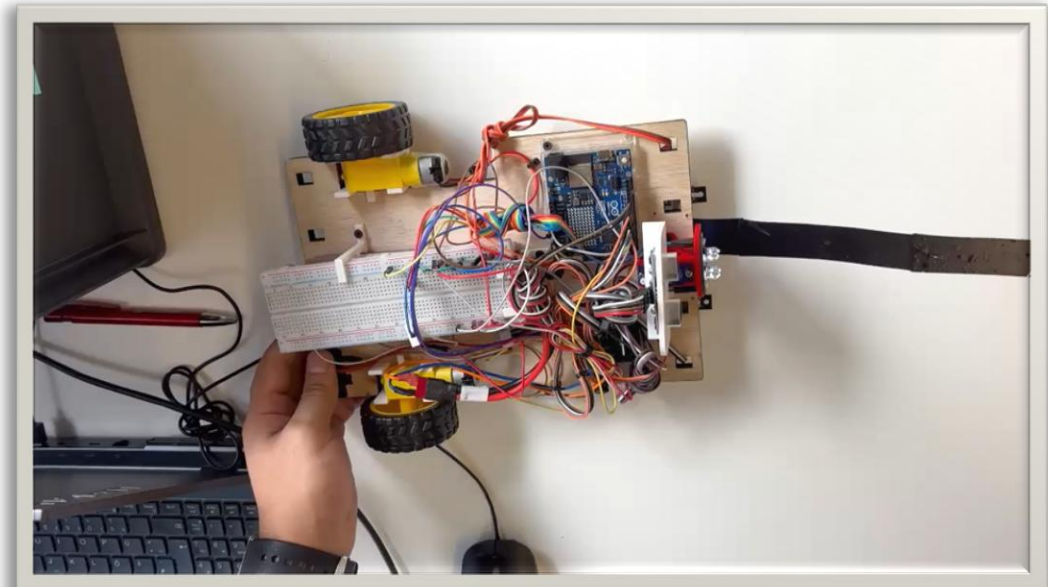


- wiring connectins.
- Arduino controls
sensors and motors
- Initial Code testing

https://www.tinkercad.com/things/dzwxTrQqEL3-swanky-snaget/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard&sharecode=xYkdW4VoKbrRX9IXlcNGk5c59ltT1bGMOaSqg_Oswtw

LANE FOLLOWING

```
if (leftIR == 0 && rightIR == 0) {  
    moveForward(speedA, speedB);  
    analogWrite(enA, forwardSpeed);  
    analogWrite(enB, forwardSpeed);  
    forward();  
} else if (leftIR == 1 && rightIR == 0) {  
    leftBackward(60);  
    rightForward(80);  
    curveLeft();  
    Stop(); delay(30);  
  
} else if (leftIR == 0 && rightIR == 1) {  
    leftForward(80);  
    rightBackward(60);  
    curveRight();  
    Stop(); delay(30);  
} else {  
    Stop();  
    delay(100);  
}  
}
```

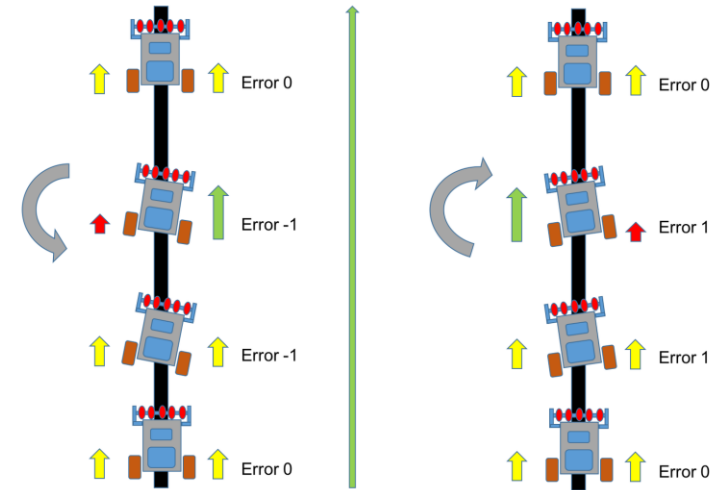


```
void loop() {  
    int leftIR = digitalRead(L_S);  
    int rightIR = digitalRead(R_S);
```

PID FOR OPTIMIZING SYSTEM

```
✓ float calculatePID(float currentVoltage) {  
    unsigned long now = millis();  
    float dt = (now - lastPIDTime) / 1000.0;  
    if (dt <= 0) dt = 0.01;  
  
    float error = batterySetpoint - currentVoltage;  
    integral += error * dt;  
    float derivative = (error - previousError) / dt;  
  
    float output = kp * error + ki * integral + kd * derivative;  
  
    previousError = error;  
    lastPIDTime = now;  
  
    return output;  
}  
  
✓ int adjustPWM(int basePWM, float pidCorrection) {  
    int adjustedPWM = basePWM + (int)pidCorrection;  
    if (adjustedPWM > 255) adjustedPWM = 255;  
    if (adjustedPWM < 0) adjustedPWM = 0;  
    return adjustedPWM;  
}
```

```
// PID parameters  
float batterySetpoint = 7.4; // Nominal battery voltage (adjust for your battery)  
float kp = 20.0;  
float ki = 0.1;  
float kd = 5.0;  
  
float integral = 0;  
float previousError = 0;  
unsigned long lastPIDTime = 0;  
  
float pidOutput = 0; // PID correction output
```



OBSTACLE DETECTION

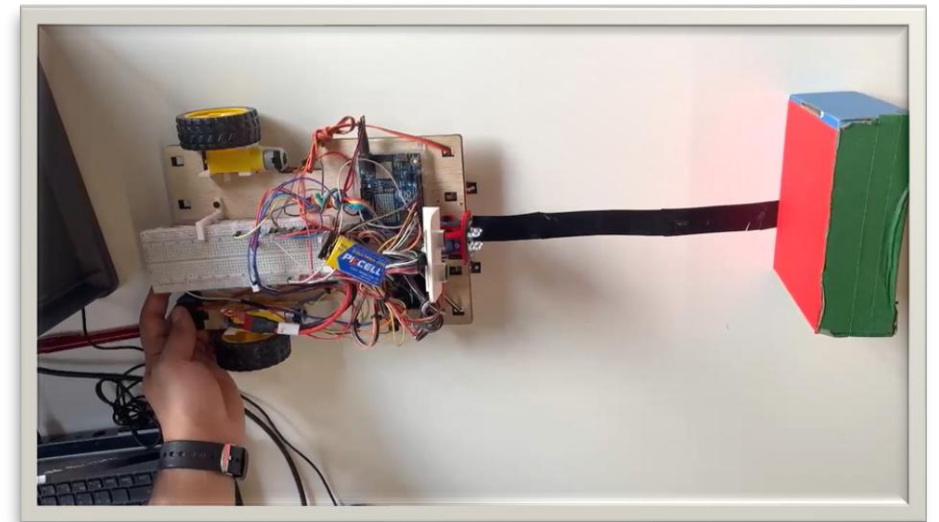
```
bool ultrasonicEnabled = true;
```

```
#define servoPin 4
```

```
#define SERVO_CENTER 70
```

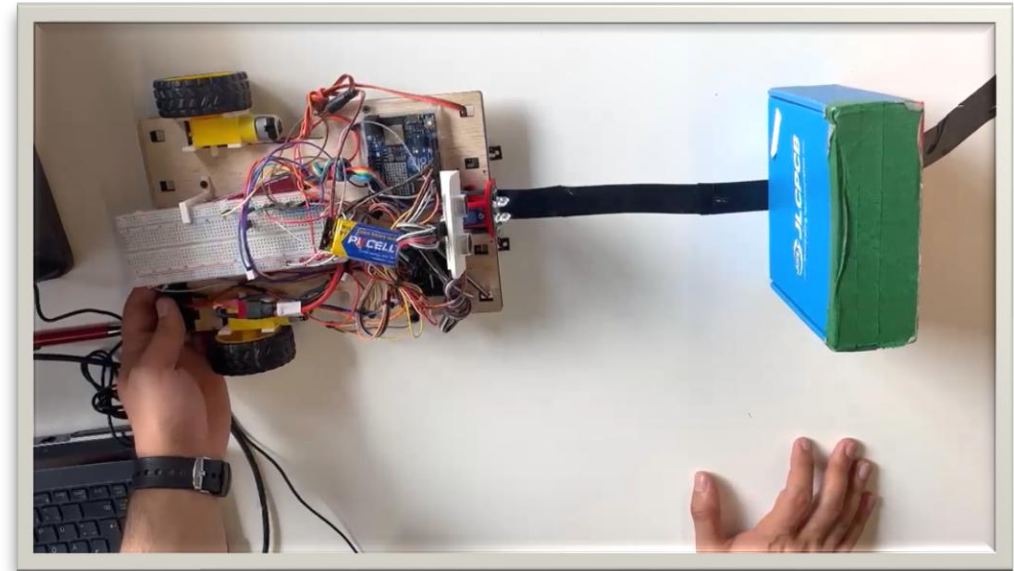
```
Servo scanServo;
```

```
✓ int getDistance() {  
    if (!ultrasonicEnabled) return 1000;  
    digitalWrite(trigPin, LOW); delayMicroseconds(2);  
    digitalWrite(trigPin, HIGH); delayMicroseconds(10);  
    digitalWrite(trigPin, LOW);  
    long duration = pulseIn(echoPin, HIGH, 20000);  
    return duration * 0.034 / 2;  
}
```



COLOR DETECTION & REACTION

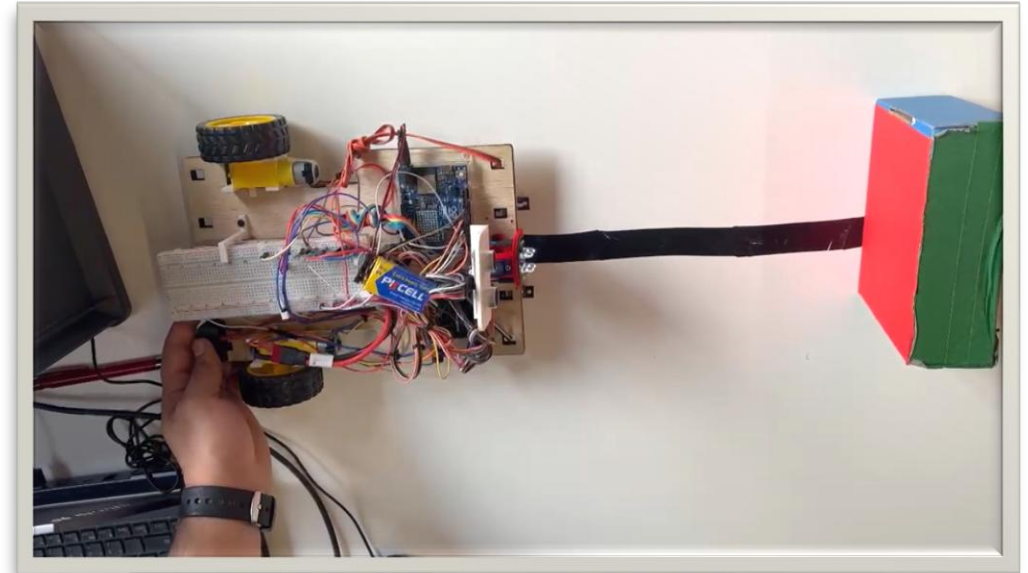
```
else if (color == "BLUE") {
    driveForward(20, 150);
    Stop(); delay(300);
    driveBackwards(30, 70);
    Stop(); delay(300);
    ultrasonicEnabled = false;
    unsigned long ignoreStart = millis();
    Serial.println("Ultrasonic disabled for 5 seconds...");
    while (millis() - ignoreStart < 5000) {
        int l = digitalRead(L_S);
        int r = digitalRead(R_S);
        if (l == 0 && r == 0) {
            analogWrite(enA, forwardSpeed);
            analogWrite(enB, forwardSpeed);
            forward();
        } else if (l == 1 && r == 0) {
            Stop(); delay(30);
            curveLeft();
        } else if (l == 0 && r == 1) {
            curveRight();
            Stop(); delay(30);
        } else {
            Stop();
            delay(100);
        }
        delay(20);
    }
    ultrasonicEnabled = true;
}
```



Blue Color Detected and Push the Object

COLOR DETECTION & REACTION

```
void handleDetectedColor(String color) {
  int colorScenarioSpeed = 80;
  if (color == "RED") {
    driveBackwards(12, colorScenarioSpeed);
    scanServo.write(180); delay(400);
    int leftDist = getDistance();
    scanServo.write(SERVO_CENTER); delay(200);
    if (leftDist > 20) {
      ultrasonicEnabled = false;
      rotate90(); delay(50);
      driveForward(30, colorScenarioSpeed); delay(200);
      rotateRight90(); delay(200);
      driveForward(20, colorScenarioSpeed);
      approachRightToLane();
      ultrasonicEnabled = true;
      Serial.println("RED: Path left is clear, performed left-right bypass.");
    } else {
      Serial.println("RED: Left not clear, only backed up and stopped.");
    }
    scanServo.write(SERVO_CENTER);
    Stop();
  }
}
```



Red Color Detected and turn 90° to avoid the Object & return back to lane

CHALLENGES & OBSERVATIONS

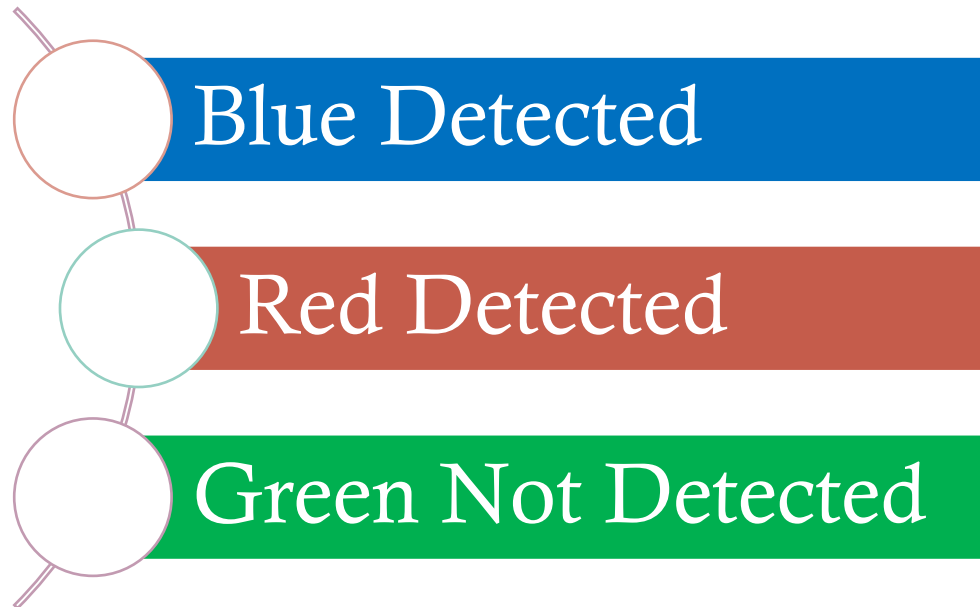


Table 1. Selectable Options

S0	S1	OUTPUT FREQUENCY SCALING (f_o)
L	L	Power down
L	H	2%
H	L	20%
H	H	100%

S2	S3	PHOTODIODE TYPE
L	L	Red
L	H	Blue
H	L	Clear (no filter)
H	H	Green

“TCS3200 Programmable Color Light-to-Frequency Converter,” Texas Advanced Optoelectronic Solutions (TAOS), [Online]. Available: <https://www.mouser.com/catalog/specsheets/tcs3200-e11.pdf>. [Accessed: Jun. 27, 2024]

THANK YOU FOR YOUR TIME 😊

QUESTIONS OR FEEDBACKS?

